

DSC 140A

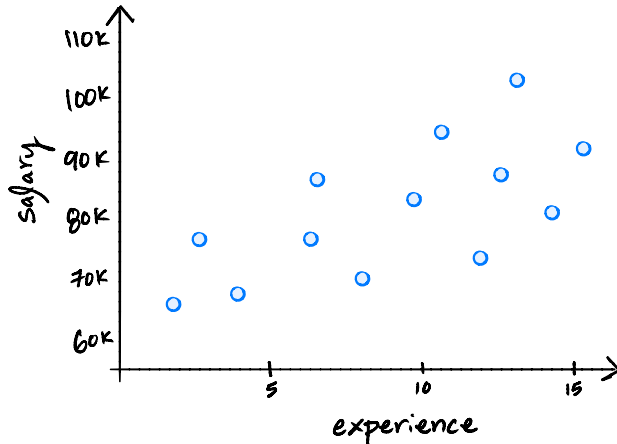
Probabilistic Modeling & Machine Learning

Lecture 16 | Part 1

Recall: Regression

Recall

- We have seen the problem of regression.



Recall

- ▶ Introduced **empirical risk minimization (ERM)**:
- ▶ Step 1: choose a **hypothesis class**
 - ▶ Let's assume we've chosen linear predictors
- ▶ Step 2: choose a **loss function**
 - ▶ Used square loss
- ▶ Step 3: minimize **expected loss (empirical risk)**
 - ▶ MSE (Mean Squared Error)

Recall: Least Squares

- ▶ Goal: fit a function of the form $H(\vec{x}; \vec{w}) = \text{Aug}(\vec{x}) \cdot \vec{w}$
- ▶ In (ordinary) least squares regression, we **minimized** the **mean squared error**:

$$\vec{w}^* = \arg \min_{\vec{w}} \frac{1}{n} \sum_{i=1}^n (H(\vec{x}^{(i)}; \vec{w}) - y_i)^2$$

- ▶ **Solution:** $\vec{w}^* = (X^T X)^{-1} X^T \vec{y}$

Observation

- ▶ This the “curve fitting” approach to regression.
- ▶ I.e., find a “line of best fit”.
- ▶ There was no consideration of the (random) process that generated the data.

Today

- ▶ Take a probabilistic approach to regression.

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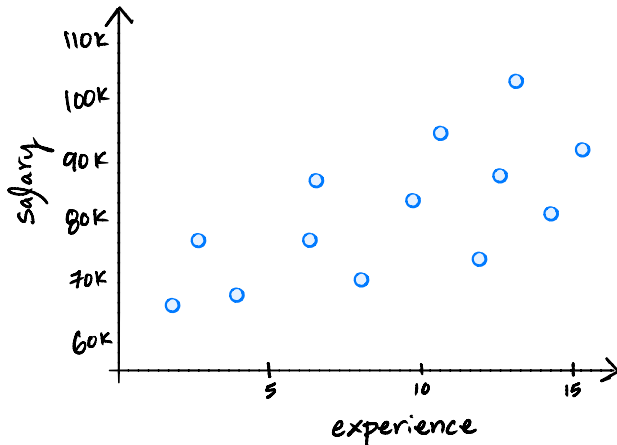
Probabilistic Modeling & Machine Learning

Lecture 16 | Part 2

Probabilistic View of Regression

Probabilistic View of Regression

- **Note:** There is **uncertainty** in the salary.

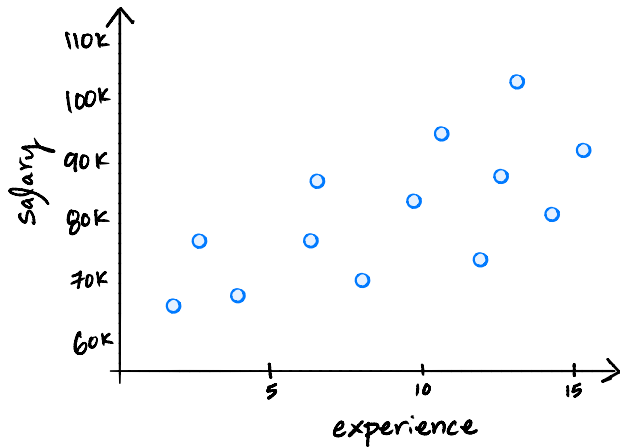


Modeling Uncertainty

- ▶ We can model this uncertainty using probability.

$$\text{Salary} = w_0 + w_1 \times (\text{Experience}) + \varepsilon$$

- ▶ Here, ε is the (random) **error**.
- ▶ What is a reasonable choice of **distribution** for ε ?



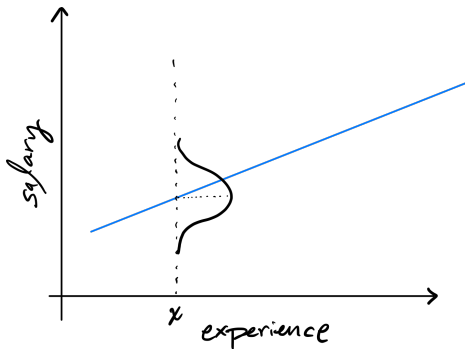
Error Distribution

- ▶ It is reasonable to assume that the error distribution is:
 - ▶ **Symmetric:** equally as likely to predict high as to predict low
 - ▶ **Centered at zero:** mean error is zero
- ▶ The **Gaussian distribution** (with mean 0) satisfies this.

Modeling Uncertainty

- Assuming a Gaussian (Normal) distribution:

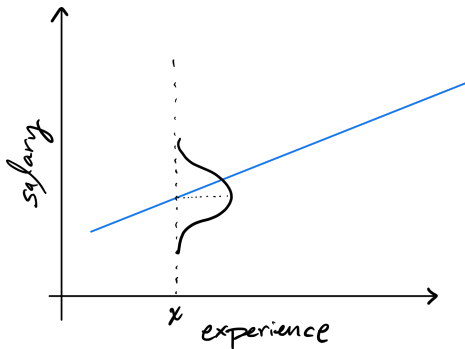
$$\text{Salary} = w_0 + w_1 \times (\text{Experience}) + \underbrace{\mathcal{N}(0, \sigma^2)}_{\varepsilon}$$



Modeling Uncertainty

- Equivalently:

$$\text{Salary} \sim \mathcal{N}(w_0 + w_1 \times \text{Experience}, \sigma^2)$$



In General

- ▶ In general:

$$Y \sim \mathcal{N}(\text{Aug}(\vec{x}) \cdot \vec{w}, \sigma^2)$$

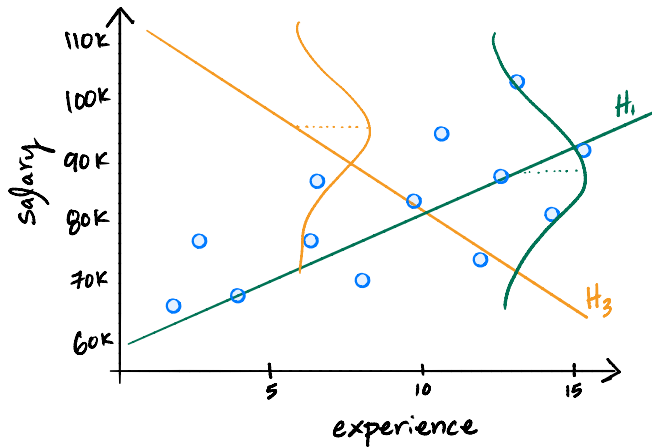
- ▶ That is: for any feature vector $\vec{x}$, the target Y is drawn from a Gaussian centered at $\text{Aug}(\vec{x}) \cdot \vec{w}$.

Estimating Parameters

- ▶ We assume the model:

$$\text{Salary} \sim \mathcal{N}(w_0 + w_1 \times \text{Experience}, \sigma^2)$$

- ▶ Given some data, what parameters generated it?
 - ▶ What were w_0 , w_1 , σ ?
- ▶ **Estimate** them with maximum likelihood?



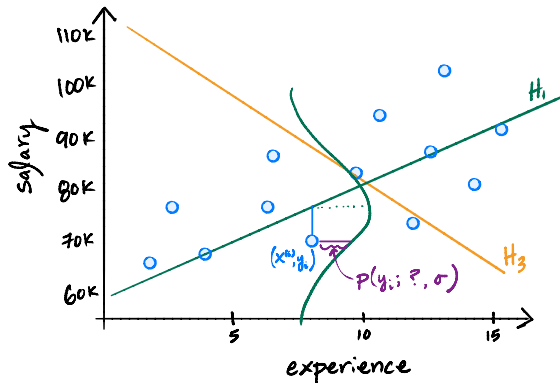
Likelihood

- ▶ Let $p(y; \mu, \sigma)$ be the Gaussian pdf:

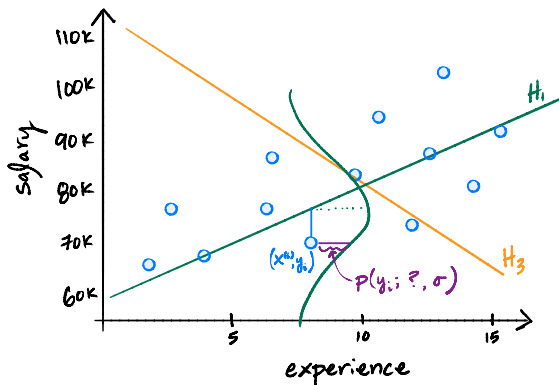
$$p(y; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(y-\mu)^2/(2\sigma^2)}$$

- ▶ We observe a data set $\{(\vec{x}^{(i)}, y_i)\}$.
- ▶ What is the likelihood of a choice of parameters $\vec{w}, \sigma$, with respect to the data?

Likelihood wrt a Point



Likelihood wrt a Point



- $p(y_i; w_0 + w_1 x^{(i)}, \sigma)$ measures likelihood with respect to $(x^{(i)}, y_i)$.

Likelihood

- In general,

$$p(y_i; \text{Aug}(\vec{x}^{(i)}) \cdot \vec{w}, \sigma)$$

measures likelihood with respect to single data point $(\vec{x}^{(i)}, y_i)$.

- Likelihood with respect to data set:

$$L(\vec{w}, \sigma) = \prod_{i=1}^n p(y_i; \text{Aug}(\vec{x}^{(i)}) \cdot \vec{w}, \sigma)$$

Log-Likelihood

Compute the log-likelihood from
 $\prod_{i=1}^n p(y_i; \text{Aug}(\vec{x}^{(i)}) \cdot \vec{w}, \sigma).$

Log-Likelihood

- The log-likelihood is:

$$\tilde{L}(\vec{w}, \sigma) = -\frac{1}{2\sigma^2} \sum_{i=1}^n (\text{Aug}(\vec{x}^{(i)}) \cdot \vec{w} - y_i)^2 + \frac{n}{2} \ln \frac{1}{\sigma^2} - \frac{n}{2} \ln(2\pi)$$

- We want to **maximize** this quantity.

Claim 1

$$\begin{aligned} & \arg \max_{\vec{w}} \left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (\text{Aug}(\vec{x}^{(i)}) \cdot \vec{w} - y_i)^2 + \frac{n}{2} \ln \frac{1}{\sigma^2} - \frac{n}{2} \ln(2\pi) \right] \\ & = \\ & \arg \max_{\vec{w}} \left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (\text{Aug}(\vec{x}^{(i)}) \cdot \vec{w} - y_i)^2 \right] \end{aligned}$$

Claim 2

$$\begin{aligned} & \arg \max_{\vec{w}} \left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (\text{Aug}(\vec{x}^{(i)}) \cdot \vec{w} - y_i)^2 \right] \\ &= \\ & \arg \max_{\vec{w}} \left[-\frac{1}{n} \sum_{i=1}^n (\text{Aug}(\vec{x}^{(i)}) \cdot \vec{w} - y_i)^2 \right] \end{aligned}$$

Claim 3

$$\begin{aligned} & \arg \max_{\vec{w}} \left[-\frac{1}{n} \sum_{i=1}^n (\text{Aug}(\vec{x}^{(i)}) \cdot \vec{w} - y_i)^2 \right] \\ &= \\ & \arg \min_{\vec{w}} \left[\frac{1}{n} \sum_{i=1}^n (\text{Aug}(\vec{x}^{(i)}) \cdot \vec{w} - y_i)^2 \right] \end{aligned}$$

- That is, minimize the **mean squared error**.

Main Idea

Maximizing the likelihood of $\vec{w}$ with respect to the data (assuming Gaussian error term) is **equivalent** to minimizing mean squared error.

Solution

- ▶ The maximum likelihood estimate for $\vec{w}$ is therefore:

$$\vec{w}^* = (X^T X)^{-1} X^T \vec{y}$$

- ▶ That is, the exact same as we obtained by empirical risk minimization with the square loss.

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Probabilistic Modeling & Machine Learning

Lecture 16 | Part 3

A Probabilistic View of Regularization

Recall: Ridge Regression

- ▶ In **ridge regression**, we added a regularization term: $\|\vec{w}\|^2$.

$$\vec{w}^* = \arg \min_{\vec{w}} \frac{1}{n} \sum_{i=1}^n (H(\vec{x}^{(i)}; \vec{w}) - y_i)^2 + \lambda \|\vec{w}\|^2$$

- ▶ **Solution:** $\vec{w}^* = (X^T X + n\lambda I)^{-1} X^T \vec{y}$
- ▶ Helps control overfitting.

Probabilistic View

- ▶ Regularization term $\|\vec{w}\|^2$ was motivated by observing that $\|\vec{w}\|$ tends to be large when overfitting.
- ▶ Now: motivate same term, probabilistically.
- ▶ Will adopt a **Bayesian** perspective.

A Prior on Weights

- ▶ Imagine we have yet to see the data.
- ▶ There is no reason to believe that a given weight w_i is positive or negative.
- ▶ We believe it is more likely to be small (close to zero) than large.

A Prior on Weights

- ▶ This **prior belief** is captured by assuming:

$$w_i \sim \mathcal{N}(0, s^2)$$

- ▶ Note that in truth, w_i is **not** random.
- ▶ We are adopting a **Bayesian** view of probability; it expresses level of belief.

A Prior on Weights

- If each weight has distribution $\mathcal{N}(0, s^2)$, then:

$$\vec{w} \sim \mathcal{N}(\vec{0}, s^2 \cdot I)$$

- That is, the distribution of $\vec{w}$ has density:

$$p_{\vec{w}}(\vec{w}) = \frac{1}{(2\pi s^2)^{d/2}} e^{-\frac{1}{2} \frac{\|\vec{w}-\vec{0}\|^2}{s^2}}$$

Distribution of $\vec{w}$

- ▶ Using Bayes' Rule:

$$p_{\vec{w}}(\vec{w} \mid \vec{x}, y) \propto p_y(y \mid \vec{w}, \vec{x})p_{\vec{w}}(\vec{w})$$

- ▶ What is the most probable value of $\vec{w}$?

$$\begin{aligned}
\arg \max_{\vec{w}} [p_{\vec{w}}(\vec{w} \mid \vec{x}, y)] &= \arg \max_{\vec{w}} [p_y(y \mid \vec{w}, \vec{x}) p_{\vec{w}}(\vec{w})] \\
&= \arg \max_{\vec{w}} \ln [p_y(y \mid \vec{w}, \vec{x}) p_{\vec{w}}(\vec{w})] \\
&= \arg \max_{\vec{w}} [\ln p_y(y \mid \vec{w}, \vec{x}) + \ln p_{\vec{w}}(\vec{w})] \\
&= \arg \min_{\vec{w}} [-\ln p_y(y \mid \vec{w}, \vec{x}) - \ln p_{\vec{w}}(\vec{w})] \\
&= \arg \min_{\vec{w}} [\text{MSE}(\vec{w}) - \ln p_{\vec{w}}(\vec{w})]
\end{aligned}$$

Deriving the Regularizer

► Since

$$p_{\vec{w}}(\vec{w}) = \frac{1}{(2\pi s^2)^{d/2}} e^{-\frac{1}{2} \frac{\|\vec{w} - \vec{0}\|^2}{s^2}}$$

we have:

$$-\ln p_{\vec{w}}(\vec{w}) = c + \frac{1}{2s^2} \|\vec{w}\|^2$$

► So

$$\arg \min_{\vec{w}} [\text{MSE}(\vec{w}) - \ln p_{\vec{w}}(\vec{w})] = \arg \min_{\vec{w}} \left[\text{MSE}(\vec{w}) + \underbrace{\frac{1}{2s^2}}_{\lambda} \|\vec{w}\|^2 \right]$$

Main Idea

Placing a $\mathcal{N}(0, s^2)$ prior on each weight and maximizing $p_{\vec{w}}(\vec{w} \mid \vec{x}, y)$ is equivalent to minimizing the $\|\vec{w}\|^2$ -regularized mean squared error (**ridge regression**).